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Professional Communication and Technical Writing

AAS105

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 1 | **Credits:** 2 | **L-T-P:** 2-0-0

Scheme: AKTU B.Tech 1st year, 2026-27 onwards

UNIT 1 Professional Communication and Business Writing

UNIT 2 Grammar, Vocabulary and Editing

UNIT 3 Public Speaking, Presentations and AI-Assisted Communication

UNIT 4 Emotional Intelligence and Critical Thinking

UNIT 5 Leadership, Teamwork and Well-being

Original revision notes written for the AKTU syllabus. This is a first-year course common to all branches. The same course is taught as AAS205 in semester 2 for the other group of students. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: processes, formats and comparison tables.
3. **Worked examples**: ready formats (email, notice, agenda, minutes, cover letter) and corrected sentences.
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (2 hours):** Unit 1 (communication process, barriers, 7 Cs, email, notice, agenda, minutes, report, proposal, resume, cover letter, interview, LinkedIn). Practise writing one email and one set of minutes from memory.
- **Afternoon (2 hours):** Unit 2 (parts of speech, common errors, subject-verb agreement, tenses, articles, word formation, tone) and Unit 3 (speaking, non-verbal communication, presentations, slides, elevator pitch, AI tools).
- **Evening (1 to 2 hours):** Units 4 and 5 (emotional intelligence, critical thinking, leadership, feedback, teamwork, stress), then the "Last-hour checklist".
- **In the exam:** use the correct format with all its parts (they carry marks), keep sentences short, use headings and bullet points, and give an example for every theory point.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions

(one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit. Internal assessment carries another 30 marks.

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UNIT 1 Professional Communication and Business Writing

✍️ 2-mark definitions

- **Communication:** the process of sharing information, ideas and feelings between a sender and a receiver so that both understand the same meaning.
- **Technical communication:** communication of specialised information (procedures, data, designs) to a specific audience for a specific purpose, in a clear, accurate and concise form.
- **Feedback:** the receiver's response that tells the sender whether the message was understood. It completes the communication loop.
- **Noise:** anything that disturbs or distorts the message (physical noise, poor network, jargon, emotions).
- **Formal and informal channels:** formal channels follow the official structure (memos, reports, meetings); informal ones are unofficial (the "grapevine").
- **Netiquette:** good manners in online communication: clear subject lines, polite tone, no all-capitals ("shouting"), replying in time, respecting privacy.
- **ATS (applicant tracking system):** software that companies use to screen resumes by keywords and format before a person reads them.

💡 Core concepts

Communication process: sender → idea → **encoding** (words, symbols) → **channel** (email, speech, video) → **decoding** by the receiver → understanding → **feedback** to the sender. Noise can enter at any stage.

Directions of communication in an organisation: downward (manager to team: instructions), upward (team to manager: reports, suggestions), horizontal (between peers: coordination), diagonal (across departments and levels).

Barriers and remedies:

Barrier	Example	Remedy
Physical	Noise, distance, bad network	Quiet place, good equipment
Semantic (language)	Jargon, ambiguous words	Simple words, define terms
Psychological	Anger, anxiety, prejudice	Stay calm, listen with an open mind
Cultural	Different customs and gestures	Learn about the audience
Organisational	Too many levels, rigid rules	Open-door policy, clear channels
Perceptual	Different assumptions	Ask questions, confirm understanding

The 7 Cs of communication:

1. **Clear:** one main idea, simple words.
2. **Concise:** no unnecessary words.

3. **Concrete:** specific facts and figures, not vague claims.
4. **Correct:** accurate facts, grammar and spelling.
5. **Coherent:** logical flow, connected points.
6. **Complete:** all needed information (who, what, when, where, why, how).
7. **Courteous:** polite and respectful (some lists use "considerate" for the reader's point of view).

Business documents (parts to remember):

- **Email:** subject line, salutation, body (purpose first, details, action needed), closing, signature with name and contact.
- **Notice:** organisation name, the word NOTICE, date, heading, body (what, when, where, who), signature and designation. Usually in a box, about 50 words.
- **Agenda:** the list of items for a meeting, sent in advance: date, time, venue, items in order, ending with "any other business".
- **Minutes:** the official record of a meeting: date, time, venue, chairperson, members present and absent, items discussed, decisions, action points (who and by when), time of closing, signature.
- **Report (formal):** title page, abstract or executive summary, table of contents, introduction, method, findings, discussion, conclusions, recommendations, references, appendices.
- **Proposal:** title, background (problem), objectives, method, timeline, budget, expected outcomes.

Resume, CV and cover letter:

Feature	Resume	CV (curriculum vitae)
Length	1 page (freshers) to 2 pages	2 or more pages
Focus	Skills and achievements for one job	Complete academic and research record
Used for	Jobs and internships	Academics, research, abroad applications

- **Resume sections:** name and contact (email, phone, LinkedIn, GitHub), career objective or summary, education, technical skills, projects, internships, achievements, certifications, extracurriculars.
- **ATS-friendly tips:** simple layout without tables or graphics, standard headings, keywords from the job description, PDF or DOCX as asked, action verbs (designed, built, led) with numbers ("reduced load time by 30%").
- **Cover letter:** a one-page letter that introduces you, says which post you want, shows why you fit (two or three matching strengths with proof) and asks for an interview.

Interviews: types (personal, telephonic, video, panel, group discussion, technical, HR). Prepare by researching the company, revising your resume and projects, and practising common questions using the **STAR** method (situation, task, action, result).

LinkedIn profile: professional photo, clear headline ("B.Tech CSE student | Python | Machine learning"), summary, education, skills, projects, certifications, recommendations; connect and post thoughtfully.

Worked examples

Example 1 (email).

```
To: training.cell@college.edu
Subject: Request for permission to attend IoT workshop, 12 to 14 March

Dear Sir/Madam,

I am Asha Verma, B.Tech CSE first year (Roll No. 2601xxxx). I request permission
to attend a three-day IoT workshop at IIT Kanpur from 12 to 14 March. The workshop
covers sensors, ESP32 programming and cloud dashboards, which will help with my
semester project.

I will submit a short report on my return. The brochure is attached.

Thank you.

Regards,
Asha Verma
B.Tech CSE, Section A | +91 9xxxxxxxxx
```

Example 2 (notice).

```
            ABC INSTITUTE OF TECHNOLOGY, LUCKNOW
                         NOTICE
5 August 2026
                    CODING CONTEST "BYTE-26"
All first-year students are informed that the Computer Society will hold a coding
contest on 20 August 2026, 10 am to 1 pm, in Lab 3. Interested students may
register with the undersigned by 15 August. Certificates for all participants.
                                                        Rahul Singh
                                                        Secretary, Computer Society
```

Example 3 (minutes, key part). "Item 2: Budget for the tech fest. The treasurer presented the estimate of ₹1.2 lakh. Decision: approved with a 10% cut in the decoration budget. Action: treasurer to revise the budget and circulate it by 10 August."

Key points to remember

- Process: sender, encoding, channel, decoding, receiver, feedback; noise anywhere.
- Barriers: physical, semantic, psychological, cultural, organisational, perceptual.
- 7 Cs: clear, concise, concrete, correct, coherent, complete, courteous.
- Agenda is before the meeting; minutes are the record after it.
- Resume vs CV; ATS keywords; STAR method in interviews.

⭐ Important questions

1. Explain the communication process with a diagram. Explain the barriers to communication and how to overcome them.
2. Explain the 7 Cs of effective communication with examples.
3. Write an email, a notice, an agenda and minutes of a meeting on a given situation.
4. Explain the structure of a formal report and a proposal.

5. Prepare a resume and a cover letter for an internship. How do you prepare for an interview and build a LinkedIn profile?

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UNIT 2 Grammar, Vocabulary and Editing

✍️ 2-mark definitions

- **Parts of speech:** noun, pronoun, verb, adjective, adverb, preposition, conjunction and interjection (articles are a kind of adjective, called determiners).
- **Subject-verb agreement:** a singular subject takes a singular verb and a plural subject takes a plural verb ("The list of items **is** ready").
- **Tense:** the form of a verb that shows time: present, past and future, each in simple, continuous, perfect and perfect continuous forms (12 in all).
- **Articles:** "a" before a consonant sound, "an" before a vowel sound, "the" for a specific or already known thing.
- **Word formation:** making new words with prefixes (un-, re-, pre-), suffixes (-tion, -ment, -able), compounding (software, keyboard) and conversion (to email).
- **Tone:** the attitude the writing shows (formal, friendly, neutral, urgent). Professional writing is usually formal, positive and polite.
- **Editing and proofreading:** editing improves content, structure and clarity; proofreading is the final check for spelling, grammar and punctuation errors.

💡 Core concepts

Subject-verb agreement rules:

1. Two subjects joined by "and" take a plural verb: "Ravi and Asha **are** in the lab."
2. With "either...or" or "neither...nor", the verb agrees with the nearer subject: "Neither the teacher nor the students **were** late."
3. "Each", "every", "everyone", "nobody" take a singular verb: "Each of the students **has** a laptop."
4. A phrase between subject and verb does not change the verb: "The quality of the components **is** good."
5. Collective nouns usually take a singular verb when acting as one group: "The team **has** won."

Tense uses:

Tense	Use	Example
Simple present	Facts, habits	Water boils at 100 °C.
Present continuous	Happening now	I am writing the report.
Present perfect	Past action with present result	We have completed the testing.
Simple past	Finished at a known time	The meeting ended at 5 pm.
Past perfect	Earlier of two past actions	The train had left before we arrived.
Simple future	Future plans	The results will be declared on Monday.

- Write methods and results of experiments in the **past tense**; facts and conclusions in the **present tense**.

Common errors (wrong → right):

Wrong	Right	Rule
He don't know.	He doesn't know.	Singular subject
I am having two laptops.	I have two laptops.	"Have" (possess) is not used in continuous form
She is an university student.	She is a university student.	"University" starts with a "yoo" sound
One of my friend is here.	One of my friends is here.	"One of" + plural noun + singular verb
Discuss about the problem.	Discuss the problem.	"Discuss" takes no "about"
I look forward to meet you.	I look forward to meeting you.	"To" here is a preposition
Please revert back.	Please reply.	"Revert" means go back; "back" is extra
Its a good idea.	It's a good idea.	It's = it is; its = belonging to it

Commonly confused words: affect (verb) / effect (noun); accept / except; principal / principle; stationary (not moving) / stationery (paper); loose / lose; complement / compliment; advice (noun) / advise (verb).

Tone and plain style: use active voice where possible ("We tested the circuit" rather than "The circuit was tested by us"), short sentences (15 to 20 words), positive wording ("Please submit by Friday" rather than "Do not be late"), and no slang in formal writing.

Key points to remember

- 8 parts of speech; 12 tenses.
- Either/neither: verb agrees with the nearer subject; each/every: singular.
- a/an by sound, not spelling (an hour, a university).
- Experiments in past tense; facts in present tense.
- Know the confused-word pairs and the common errors table.

⭐ Important questions

1. Explain the rules of subject-verb agreement with examples.
2. Explain the uses of tenses in technical writing.
3. Correct the given sentences and explain the errors.
4. Explain word formation with prefixes, suffixes and compounding; give commonly confused words.
5. What is tone in professional writing? Explain editing and proofreading techniques.

UNIT 3 Public Speaking, Presentations and AI-Assisted Communication

✍️ 2-mark definitions

- **Public speaking:** speaking to a group in a planned, organised way to inform, persuade or inspire.
- **Modes of delivery: impromptu** (no preparation), **extempore** (prepared outline, spoken naturally), **manuscript** (read word for word) and **memorised**. Extempore is best for most presentations.
- **Verbal communication:** the use of words, spoken or written.
- **Non-verbal communication:** messages sent without words: **kinesics** (body language, gestures, posture), **facial expression and eye contact**, **proxemics** (use of space), **paralanguage** (tone, pitch, pace, pauses), **chronemics** (use of time) and appearance.
- **Abhinaya (Natyashastra):** Bharata's Natyashastra describes four kinds of expression: **angika** (body movement), **vachika** (speech and voice), **aharya** (costume and appearance) and **sattvika** (inner emotion shown naturally). These map closely to the parts of modern non-verbal communication.
- **Elevator pitch:** a short (30 to 60 second) introduction of yourself or an idea that makes the listener want to know more.
- **AI-assisted communication:** using AI tools to draft, edit, translate, summarise or design messages, with a human checking and taking responsibility.

💡 Core concepts

Preparing a presentation:

1. Know the **purpose** and the **audience** (their level and interest).
2. **Research** and select three to five key points.
3. **Structure:** opening (hook: a question, fact or story), body (key points with evidence), closing (summary and call to action), then questions.
4. **Design slides.**
5. **Rehearse** aloud and time it.
6. **Deliver** with eye contact, open posture, varied voice and natural pauses.
7. **Handle questions:** listen fully, repeat the question, answer briefly, and admit when you do not know.

Slide design rules:

- One idea per slide; a clear title.
- About **6 lines of 6 words** at most (the 6 × 6 rule); no full paragraphs.
- Large font (titles 32 pt or more, text 24 pt or more), high contrast, at most two fonts.
- Use charts, diagrams and images instead of text where possible; label axes and give sources.
- Consistent colours and layout; few animations.

Elevator pitch structure: who you are → what you do or are working on → what makes it valuable or different → what you want (a call to action). Example: "I'm Rohan, a first-year CSE student. I built a

Python app that tracks hostel mess waste and has cut it by 15% in a month. I'm looking for a summer internship where I can work on data dashboards."

Managing stage fear: prepare well, rehearse, breathe slowly, focus on the message rather than yourself, start with a well-practised opening, and look at friendly faces.

AI tools in communication:

- **Uses:** grammar and style checking, drafting emails and reports, summarising long documents, translation, generating slide outlines and images, speech practice with feedback.
- **Prompting for communication:** state the audience, purpose, tone, length and format ("Rewrite this email to my professor in a polite, formal tone in under 120 words").
- **Ethics:** check every fact; do not present AI work as your own where it is not allowed; disclose its use when required; never share confidential or personal data; keep your own voice.

Key points to remember

- Delivery modes: impromptu, extempore, manuscript, memorised.
- Non-verbal: kinesics, proxemics, paralanguage, chronemics, eye contact, appearance.
- Natyashastra abhinaya: angika, vachika, aharya, sattvika.
- Presentation: hook, 3 to 5 points, summary, call to action; 6 × 6 slide rule.
- AI: draft and refine, but verify and disclose.

⭐ Important questions

1. Explain the modes of delivery and the qualities of an effective speaker.
2. Explain verbal and non-verbal communication; relate non-verbal communication to the abhinaya of the Natyashastra.
3. Explain how to plan and deliver an effective presentation, with slide design rules.
4. What is an elevator pitch? Prepare one for yourself.
5. Discuss the use of AI tools in professional communication, with prompting tips and ethical concerns.

UNIT 4 Emotional Intelligence and Critical Thinking

✍️ 2-mark definitions

- **Emotional intelligence (EI):** the ability to recognise, understand and manage your own emotions and to recognise and influence the emotions of others. Popularised by Daniel Goleman.
- **Goleman's five components:** self-awareness, self-regulation, motivation, empathy and social skills.
- **Empathy:** understanding another person's feelings and point of view.
- **Critical thinking:** careful, logical analysis of information and arguments to reach a sound judgement, rather than accepting claims at face value.
- **Viveka:** in Indian thought, the faculty of discrimination between right and wrong, real and unreal, lasting and passing. It is close to wise, reflective judgement.
- **Nyaya method:** the Nyaya school's method of valid reasoning using pramanas (perception, inference, comparison, testimony) and a five-step argument.
- **Cognitive bias:** a systematic error in thinking, such as confirmation bias (favouring information that supports what we already believe).

💡 Core concepts

EI in the Bhagavad Gita:

- **Sthitaprajna** (Chapter 2): a person of steady wisdom who is not shaken by pleasure or pain, success or failure. This is self-regulation.
- The chain from attachment to desire, anger, confusion and loss of judgement (Chapter 2, verses 62 and 63) explains how unmanaged emotions lead to poor decisions.
- **Samatvam** (equanimity) and doing one's duty without anxiety about results reduce stress and improve focus.

Steps of critical thinking:

1. Identify the problem or claim.
2. Gather relevant information and evidence.
3. Check the sources (reliable, current, unbiased?).
4. Analyse the arguments and look for assumptions and biases.
5. Consider alternatives and consequences.
6. Reach a reasoned conclusion and be ready to revise it.

Nyaya five-step argument (panchavayava):

Step	Name	Example
1	Pratijna (claim)	The hill has fire.
2	Hetu (reason)	Because it has smoke.
3	Udaharana (example)	Wherever there is smoke there is fire, as in a kitchen.

Step	Name	Example
4	Upanaya (application)	This hill has smoke.
5	Nigamana (conclusion)	Therefore the hill has fire.

Value-based decision making: list the options; check each against facts, rules and values (honesty, fairness, safety, the welfare of others); consider who is affected; choose and take responsibility. For engineers, safety and the public good come first.

Key points to remember

- Goleman: self-awareness, self-regulation, motivation, empathy, social skills.
- Gita: sthitaprajna, samatvam; the anger chain of verses 2.62 to 63.
- Critical thinking steps; watch for confirmation bias.
- Nyaya five steps: pratijna, hetu, udaharana, upanaya, nigamana.
- Viveka = discriminating judgement.

⭐ Important questions

1. What is emotional intelligence? Explain Goleman's five components with workplace examples.
2. Explain the teachings of the Bhagavad Gita that relate to emotional intelligence.
3. What is critical thinking? Explain its steps and common cognitive biases.
4. Explain the Nyaya method of reasoning with the five-step argument.
5. Explain viveka and value-based decision making for engineers with an example.

UNIT 5 Leadership, Teamwork and Well-being

✍️ 2-mark definitions

- **Leadership:** the ability to guide and influence people towards a shared goal.
- **Autocratic leadership:** the leader decides alone; fast but can lower morale.
- **Democratic (participative) leadership:** the leader involves the team in decisions; better commitment, slower decisions.
- **Laissez-faire leadership:** the leader gives freedom with little direction; works with skilled, motivated teams.
- **Transformational leadership:** the leader inspires with a vision and helps people grow. **Transactional leadership:** works through rewards and penalties for meeting targets. **Servant leadership:** the leader puts the team's needs first.
- **Active listening:** giving full attention, not interrupting, asking questions and paraphrasing to confirm understanding.
- **Constructive feedback:** specific, timely feedback on behaviour (not the person) that helps improvement.
- **Stress:** the body's and mind's response to pressure; some stress motivates (eustress), too much harms (distress).
- **Yama and niyama:** the first two limbs of Patanjali's yoga: ethical restraints (yama: ahimsa, satya, asteya, brahmacharya, aparigraha) and personal observances (niyama: shaucha, santosha, tapas, svadhyaya, ishvara pranidhana).

💡 Core concepts

Leadership styles:

Style	Decision making	Good for	Risk
Autocratic	Leader alone	Emergencies, new teams	Low morale
Democratic	With the team	Creative, planning work	Slow
Laissez-faire	Team decides	Expert teams	Lack of direction
Transformational	Vision-led	Change and innovation	Depends on the leader
Servant	Team needs first	Long-term trust	Seen as weak if overdone

Giving feedback (SBI model): describe the **situation**, the **behaviour** you saw and its **impact**, then suggest an improvement. Example: "In yesterday's review (situation), you explained the circuit with a clear diagram (behaviour), which helped the client understand quickly (impact). Next time, add the cost estimate too."

Receiving feedback: listen without defending, ask for examples, thank the person, and act on it.

Tuckman's stages of team development: forming (getting to know each other) → storming (conflicts appear) → norming (rules and roles settle) → performing (the team works well) → adjourning (the task

ends).

Qualities of a good team: a shared goal, clear roles, trust, open communication, respect for differences, and shared accountability.

Stress management: plan and prioritise (to-do lists, the urgent-important matrix), break big tasks into parts, sleep 7 to 8 hours, exercise, practise yoga and pranayama, limit screen time, talk to friends, mentors or counsellors, and keep realistic goals.

Lessons from the Panchatantra: stories by Vishnu Sharma teaching practical wisdom through animal fables, for example:

- **The monkey and the crocodile:** presence of mind in a crisis.
- **The lion and the hare:** intelligence beats strength.
- **The tortoise and the geese:** failing to control one's speech brings disaster (the tortoise opens its mouth and falls).
- **The four friends (crow, mouse, tortoise, deer):** teamwork and loyalty help everyone survive.

Key points to remember

- Styles: autocratic, democratic, laissez-faire, transformational, transactional, servant.
- Active listening; SBI feedback model.
- Tuckman: forming, storming, norming, performing, adjourning.
- Yama (ahimsa, satya, asteya, brahmacharya, aparigraha); niyama (shaucha, santosha, tapas, svadhyaya, ishvara pranidhana).
- Panchatantra stories as leadership and teamwork lessons.

⭐ Important questions

1. What is leadership? Compare the different leadership styles with examples.
2. Explain active listening and how to give and receive constructive feedback.
3. Explain Tuckman's stages of team development and the qualities of an effective team.
4. What is stress? Explain stress management techniques, including yama and niyama.
5. Explain leadership and life lessons from the Panchatantra with examples.

Last-hour checklist

☐ Communication process; directions; barriers table; 7 Cs; email, notice, agenda, minutes, report and proposal formats; resume vs CV; ATS tips; STAR; LinkedIn.

☐ Parts of speech; subject-verb agreement rules; tense table; common errors table; confused words; active voice and tone.

☐ Delivery modes; non-verbal types; abhinaya (angika, vachika, aharya, sattvika); presentation steps; 6 × 6 rule; elevator pitch; AI tools and ethics.

☐ Goleman's five; Gita (sthitaprajna, 2.62 to 63); critical thinking steps; Nyaya five steps; viveka; value-based decisions.

☐ Leadership table; active listening; SBI feedback; Tuckman; stress tips; yama and niyama; Panchatantra stories.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Technical Communication: Principles and Practice	Meenakshi Raman, Sangeeta Sharma	Oxford University Press	WorldCat
2	Advanced Soft Skills	Vidushi Sharma	Pearson	WorldCat
3	Business Correspondence and Report Writing	R. C. Sharma, Krishna Mohan	McGraw Hill	WorldCat
4	Business Communication: Making Connections in a Digital World	Raymond V. Lesikar, Marie E. Flatley	McGraw Hill	WorldCat

Syllabus source: AKTU B.Tech 1st year CSE stream course structure and syllabus, 2026-27 (official PDF)

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